1. Bayesian Networks: ..... *Total: 13 marks*

(1) Dr. Johnson is a researcher who studies a special mental health disorder and its possible treatments. He has two different treatments, called A and B, that he has been testing on his patients for a long time. He has collected thousands of data records from his patients and summarised them in the table below. Let $T \in \{A, B\}$ denote the treatment a patient received and let $Y \in \{0, 1\}$ denote whether the treatment was successful ($Y = 1$).

He has also observed that his patients vary in how long they stay in negative thinking patterns. He introduces a variable C and labels the duration of negative thinking as either *short* or *long* in the records. The variable C influences which treatment a patient is given as well as the outcome of the treatment. He has also noticed that his patients differ in their level of social isolation after treatment. For that, he introduces a variable M, labelled *mild* or *severe*. The level of isolation is influenced by the treatment and in turn affects the chance of success; the treatment may also affect success directly.

The percentages in the table are the success rates within each group, and the fractions give the number of successes over the number of patients in that group. He would like to use the data to answer questions about his patients.

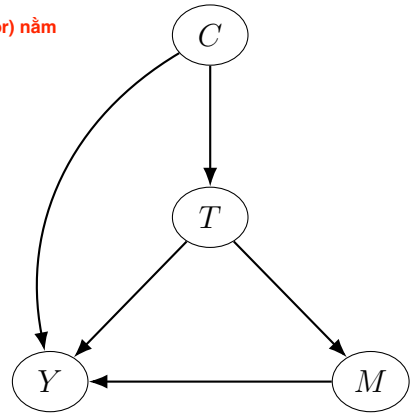
Treatment	$C = short$	$C = long$	$M = mild$	$M = severe$
A	16% (160/1000)	38% (38/100)	45% (90/200)	12% (108/900)
B	10% (5/50)	21% (126/600)	70% (105/150)	5.2% (26/500)

(5 marks) (a). Create a DAG for a Bayesian Network based on the above description.

Solution:

M ở đây là biến trung gian (Mediator) nằm trên chuỗi nhân quả T -> M -> Y

Tập nút cha Pa(Y) = {C, T, M}



(2 marks) (b). Write the joint distribution of all variables and factorise it over the DAG.

> **Solution:** $P(Y, C, M, T) = P(C)\,P(T \mid C)\,P(M \mid T)\,P(Y \mid C, T, M)$.

(2 marks) (c). Are C and M D-separated (with no variable observed)? List the paths between them and state why each is blocked or active. Does the answer change if T is observed?

> **Solution:** No, C and M are not D-separated. There are three paths between them:
>
> - $C \to T \to M$: a chain through T; T is unobserved, so this path is *active*.
> - $C \to Y \leftarrow T \to M$: Y is a collider and is unobserved, so this path is blocked.
> - $C \to Y \leftarrow M$: again a collider at Y, blocked.
>
> One active path suffices, so C and M are D-connected.
>
> If T is observed, the chain $C \to T \to M$ is blocked at T, and the two remaining paths stay blocked at the unobserved collider Y (on the second path T is now also an observed fork). Hence C and M are D-separated given T: $C \perp M \mid T$. Note that observing Y instead would *open* the collider paths.

(d). Dr. Johnson's next patient has a long duration of negative thinking ($C = long$). Use the DAG and the table to answer the following.

(2 marks) (i) If the patient is treated according to Dr. Johnson's usual practice, what is the probability that the treatment is successful, $P(Y = 1 \mid C = long)$?

(2 marks) (ii) The patient is treated and the treatment is successful, but the record of which treatment was given is lost. What is the probability that the patient received treatment A, $P(T = A \mid C = long, Y = 1)$?

> **Solution:** Both queries involve only C, T and Y; M is unobserved and is marginalised out. The relevant factors are read from the table. Among the 700 patients with $C = long$, 100 received A and 600 received B, so
>
> $$P(T = A \mid C = long) = \tfrac{100}{700} = \tfrac{1}{7}, \qquad P(T = B \mid C = long) = \tfrac{600}{700} = \tfrac{6}{7},$$
>
> and the success rates are $P(Y = 1 \mid T = A, C = long) = 0.38$ and $P(Y = 1 \mid T = B, C = long) = 0.21$. (Each of these is $\sum_M P(Y = 1 \mid C, T, M)\,P(M \mid T)$, which the table provides directly.)

(i) Marginalising over the treatment,

$$P(Y = 1 \mid C = \mathit{long}) = \sum_{t \in \{A,B\}} P(Y = 1 \mid T = t, C = \mathit{long})\, P(T = t \mid C = \mathit{long})$$
$$= 0.38 \times \tfrac{1}{7} + 0.21 \times \tfrac{6}{7} = \tfrac{0.38+1.26}{7} \approx 0.234.$$

Equivalently, $(38 + 126)/700 = 164/700$.

(ii) By Bayes' rule, using (i) as the normaliser,

$$P(T = A \mid C = \mathit{long}, Y = 1) = \frac{P(Y = 1 \mid T = A, C = \mathit{long})\, P(T = A \mid C = \mathit{long})}{P(Y = 1 \mid C = \mathit{long})}$$
$$= \frac{0.38 \times \frac{1}{7}}{0.234} = \frac{0.0543}{0.234} \approx 0.232.$$

Equivalently, 38/164. Although A has the higher success rate for long-duration patients (38% against 21%), a successful long-duration patient most likely received B, because B is given to six times as many of these patients; the prior $P(T \mid C)$ outweighs the likelihood ratio $0.38/0.21 \approx 1.8$.